The primes and the fence: an attack on Firoozbakht’s conjecture

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Leg: `write-paper` (node 20 of the `math-attack` polymer, run node) **Delivery posture**: **staged** — see §0.2. This document is not cleared for release.

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0. Posture, stated before anything else

0.1 The conjecture is open

> **Firoozbakht’s conjecture is open.** This paper does **not** prove it and does **not** refute it. No leg of the polymer that produced this paper generated any evidence about its truth in either direction, and this paper generates none either.
>
> Everything below is a report on *arguments, machine-checked formalisations, computations and citations*. Nowhere is a statement made about the conjecture’s truth value beyond the single word **open**.

The word **proved** is used in this paper under one rule and no other:

> **Proof discipline.** A statement is called **proved** only where a Lean 4 kernel checked it and `#print axioms` showed the clean line `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no `Lean.ofReduceBool`. Every such statement is listed in §4 with its verbatim Lean type. Statements established by human argument are called **established** or **argued**, never proved, and each carries its weakest premise’s tier (§1.3).

0.2 Delivery posture: staged, and why

This paper is **staged, not cleared**. Three gates stand between it and release, and none of them has passed:

gate	status	consequence for this paper
`evidence-gate`	**BLOCKED** (§8)	two live BLOCKER faults in the corpus's self-description
`citation-gate`	**NOT RUN**	no citation in this paper has been independently re-fetched and re-read by an auditor
`editorial-verdict`	**NOT RUN**	the adversarial review of this document is a separate downstream molecule

§9 lists exactly what must close before this document can be released, and §10 lists what it does not claim.

0.3 Attribution

External attribution for this work is **Noogram**. Firoozbakht's conjecture is attributed to Faride Firoozbakht (1982); §11.3 records that no primary document by Firoozbakht exists, and states what may and may not be written about that attribution.

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1. The problem, and the shape of it

1.1 Statement

Let p_n denote the n-th prime, **1-indexed**: $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$. Write $g_n := p_{n+1} - p_n$ for the n-th prime gap and $L_n := \log p_n$ (natural logarithm throughout).

> **Firoozbakht's conjecture (F).** For every $n \geq 1$,
>
> $$p_{n+1}^{1/(n+1)} \; < \; p_n^{1/n},$$
>
> equivalently: the sequence $\left(p_n^{1/n}\right)_{n \geq 1}$ is **strictly decreasing**.

The conjecture was proposed by Faride Firoozbakht in 1982 and remains open. Its earliest public record is Rivera's *Prime Puzzles* Conjecture 30 (2002) [`rivera2002conj30`, V1]; its earliest citable print record is Ribenboim, *The Little Book of Bigger Primes*, 2nd ed., p. 185 [`ribenboim2004little`, V2].

1.2 Five forms, all pointwise equivalent

The conjecture admits five natural phrasings. They are **pointwise equivalent** — index by index, with no asymptotics and no exceptional set. This is an elementary but load-bearing fact: it is what licenses proving in one form and stating in another.

form	statement	ambient
F1	$p_{n+1}^{1/(n+1)} < p_n^{1/n}$	$\mathbb{R}$, real power
F2	$p_{n+1}^{n} < p_n^{n+1}$	$\mathbb{N}$, pure integer
F3	$n \log p_{n+1} < (n+1) \log p_n$	$\mathbb{R}$
F4	$a_{n+1} < a_n$, where $a_n := (\log p_n)/n$	$\mathbb{R}$
F5	$S_n > 0$, where $S_n := (n+1)\log p_n - n \log p_{n+1}$	$\mathbb{R}$

Proof of equivalence. $p_n \geq 2 > 0$, so both sides of F1 are positive and log is strictly increasing: F1 $\iff$ $\frac{1}{n+1}\log p_{n+1} < \frac{1}{n}\log p_n$, and multiplying by $n(n+1) > 0$ gives F3. F3 $\iff$ F2 because exp is strictly increasing and $m^k = \exp(k \log m)$ for $m \geq 1$. F3 $\iff$ F4 by dividing by $n(n+1) > 0$. F3 $\iff$ F5 is the definition of S_n. $\square$

The bridges F1 $\iff$ F2, F1 $\iff$ F3 and F1 $\iff$ (the gap form below) are **proved** in the machine-checked sense of §0.1 — see §4.2.

1.3 The exact criterion: the conjecture *is* a statement about prime gaps

> **Lemma 1 (exact criterion).** For every $n \geq 1$,
>
> $$\text{F holds at } n \iff g_n < f_n, \qquad f_n := p_n^{1+1/n} - p_n = p_n\big(e^{L_n/n} - 1\big).$$

Proof. By §1.2, F at $n \iff$ $n \log p_{n+1} < (n+1)\log p_n \iff \log p_{n+1} < (1+\frac{1}{n})L_n \iff p_{n+1} < p_n^{1+1/n}$; subtract p_n. Every step is an equivalence. □

There is no side condition, no asymptotic regime, no exceptional small-n set, and no external explicit bound. Lemma 1 is the whole content of the conjecture, index by index. It is also **proved** in the machine-checked sense (`firoozbakht_iff_gap`, §4.2), in the equivalent formulation $g_n < B_n$ with $B_n := p_n(p_n^{1/n} - 1) = f_n$.

Call $\sigma_n := f_n - g_n$ the **shortfall** at n; F at $n \iff \sigma_n > 0$.

Two consequences organise everything that follows.

- **f_n is strictly decreasing in n for fixed p_n.** Hence *verifying* F needs an **upper** bound on the index $n = \pi(p_n)$, and *refuting* it needs a **lower** bound. Getting this backwards converts a verification into a refutation claim; it is the most common way to lose the thread in this subject.
- **Asymptotically** $f_n = \log^2 p_n - \log p_n - 1 + o(1)$ [`kourbatov2015upper` Thm 5, V0]. So the fence sits at height about $(\log p)^2$.

1.4 The picture

The primes walk along a road. Every step is a gap. Firoozbakht drew a fence at height roughly $(\log p)^2$ and said: **the primes never jump it — not once, not ever, all the way to infinity.**

That picture makes both branches of the attack immediately legible.

- **To prove F**, one needs an upper bound on *every* prime gap, of size $(\log p)^2$-ish, with an explicit constant strictly below 1. Nothing in analytic number theory produces a polylogarithmic gap bound at all. The best unconditional bound is $g_n \ll p_n^{0.525}$ [`bhp2001difference`, V2]; the Riemann hypothesis gives

$g_n \ll \sqrt{p_n} \log p_n$. Those are **powers of p against powers of** $\log p$. The distance is not quantitative — it is categorical.

- **To refute F**, one needs a single gap that clears the fence. At the record locus the gap reaches 92.06% of $(\log p)^2$ while the fence sits at 97.06% of the same quantity — a shortfall factor of 1.05426 (§5.1). A few percent, in a quantity that took decades of distributed computation to measure.

The proof branch is blocked by a wall. The refutation branch is *close* but not reachable. §7.4 records how this framing was itself demoted during the work.

1.5 Tiering convention

Two orthogonal ladders are used, and they are never conflated.

Claim tier — how well a *claim* is established: **L0** proved here from first principles; **L1** standard/textbook; **L2** attributed with a verified locator; **L3** heuristic, recalled, or resting on an unrefereed computational record; **[C]** additionally recomputed by a verifier script in the run. A composite claim inherits its **weakest** premise's tier.

Citation tier — how well a *citation* was checked: **V0** primary obtained and the statement read verbatim at the named locator; **V1** primary's own metadata page read; **V2** confirmed via a reliable secondary quoting the primary; **V3** bibliographic metadata only, statement at locator **not** verified. Every citation in this paper is tagged, and §11 traces each to its source-ledger row.

A V3 locator is not usable. Where this paper touches one, it says so and does not lean on it.

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2. Method

2.1 What was actually done

The attack ran as a directed polymer of specialised legs, each producing an artifact another leg had to read. In rough order:

1. **Decomposition** — reduce the conjecture to its equivalent forms and to the gap criterion (§1.2, §1.3); enumerate attack routes.
2. **Frame deliberation** — a five-perspective panel adjudicating which branch is live.

3. **Source ledger** — build the bibliography as *(citekey, verification level, precise locator, exact statement as the source writes it)*, with independent recomputation of every numeric row.
4. **Concept cards** — 31 atomic definitions/lemmas, each carrying its ledger row.
5. **Lean skeleton, then Lean probe** — state the conjecture in Lean 4 / Mathlib and attempt real proof terms.
6. **Proof attempts** — three targets, attacked as mathematics: first-failure maximality, the RH-conditional route, and the unconditional verified range.
7. **Notebooks** — three computational legs, exhaustive and lever-based.
8. **Red-team corpus** — 19 adversarial artifacts designed to slip past a naive gate.
9. **Skeptic** — an adversarial audit that re-derived every load-bearing number in independent code and re-fetched five primary sources.
10. **Evidence gate** — fail-closed adjudication.
11. **Synthesis**, then this paper.

2.2 Method commitments

Four commitments constrain everything reported here.

- **Fail-closed gates.** A gate that cannot establish its predicate returns FAIL, never "probably fine". The evidence gate returned **BLOCKED**, and §8 reports it as such.
- **The kernel is the only source of the word "proved".** §0.1.
- **The `sorry` grep is not the audit.** A raw `grep -c sorry` over the statement file returns 6, of which exactly **one** is a genuine `sorry` in code — the rest are prose discussing what remains open. Verdicts rest on the `#print axioms` whitelist, which cannot be fooled by formatting.
- **Computation corroborates or refutes; it never proves.** No finite range bears on the conjecture. A certified range is a *fidelity instrument*, not evidence.

2.3 The formal environment

Lean 4 `v4.29.0` with Mathlib `v4.29.0` (revision pinned in the run's `lake-manifest.json`). `lake build` completes with **exit status 0**, 8253 jobs. An `Audit.lean` module is part of the library, so `#print axioms` runs on *every* build and cannot silently rot. No `native_decide` appears anywhere in the sources, so `Lean.ofReduceBool` never enters the trusted base: the finite verification is kernel-checked, not compiler-trusted.

2.4 A named self-check on every result claim

The run's red-team corpus produced an artifact (`RT–12`) that builds with exit 0, is `sorry`-free, is axiom-clean, **and proves a true theorem** — where the falsehood lives in the sentence a paper would put next to it. That artifact imposes a standing requirement on this document:

> **RT-12 requirement.** For every declaration this paper names as a result, print its Lean type and confirm the conclusion is not among its hypotheses.

§4.4 discharges that requirement for all fourteen declarations named in §4.

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3. Results at a glance

question	answer
Is the conjecture **proved**?	**No.** The Lean declaration `Firoozbakht.firoozbakht` still ends in `sorry`; its axiom line contains `sorryAx`.
Is the conjecture **refuted**?	**No.** No counterexample at any scale examined.
Was anything **proved** (kernel sense)?	**Yes** — 11 new theorems plus a 26-entry prime table, including the conjecture itself for all $1 \leq n \leq 25$ and two exact refutation criteria. §4.
Was anything **established on paper**?	**Yes** — §5. Two exact criteria, a verified-range theorem at $H = 10^{20}$ under two named premises, and three dead routes with quantitative obituaries. §6.
Evidence-gate status	**BLOCKED**. §8.
Citation clearance	**Not claimed, not audited.** §9.

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4. What was PROVED — machine-checked in Lean 4 / Mathlib

Everything in this section satisfies §0.1: `lake build` exit 0, and `#print axioms` showing `[propext, Classical.choice, Quot.sound]` for each declaration.

Throughout, `nthPrime n` is the Lean 1-indexed n-th prime, `noncomputable def nthPrime (n : ℕ) : ℕ := Nat.nth Nat.Prime (n - 1)`, and `barrier n` is the exact barrier $B_n = p_n(p_n^{1/n} - 1)$ of Lemma 1.

4.1 The headline: the conjecture, proved outright, for 25 consecutive indices

```
theorem firoozbakht_of_le (n : ℕ) (h1 : 1 ≤ n) (h2 : n ≤ 25) :
    (nthPrime (n + 1) : ℝ) ^ ((1 : ℝ) / ((n : ℝ) + 1)) <
      (nthPrime n : ℝ) ^ ((1 : ℝ) / (n : ℝ))
```

This is the **literal F1 form** — real powers, not the integer surrogate — and its axiom line is clean:

```
'Firoozbakht.firoozbakht_of_le' depends on axioms: [propext,
Classical.choice, Quot.sound]
```

so it is **not circular**: it does not depend on the `sorry`-ed conjecture. It is a genuine unconditional fragment of an open problem, checked by the same kernel that checks the bridges.

Honest sizing, stated immediately and without softening. The bound $n \le 25$ is set by **compile time, not by mathematics** — the bridge lemma is uniform in n and the table extends mechanically. And $n \le 25$ is derisory next to the literature's 4×10^{18} [`kourbatov2015verification`] and 2^{64} [`visser2019verifying`]. **No finite range bears on the conjecture.** The value of this result is not its range; it is that (a) the formal method now exists, and (b) a kernel-checked range is a different kind of object from a script-checked one.

4.2 The enabling engineering: a computable bridge out of `Nat.nth`

`Nat.nth Nat.Prime` is `noncomputable` — defined by well-founded recursion over an infinite set — so `decide` cannot evaluate it and **no finite range could be machine-checked at all**. This was the formalisation's flagged main risk item.

The escape hatch is `Nat.nth_count : p n → Nat.nth p (Nat.count p n) = n`, which trades `nth` for the computable `count`. The naive move works but is quadratic in the wrong variable: a single check `Nat.count Nat.Prime 97 = 24` walks the whole range $[0, 97)$ and cost **~36 s** of kernel reduction as measured (one entry, the largest prime in the table). That does not scale.

The leg instead proves a step lemma whose cost scales with the **gap**, not the prime:

```
theorem nthPrime_succ_eq {k p q : ℕ} (hk : 1 ≤ k) (hp : nthPrime k = p)
(hq : Nat.Prime q)
    (hpq : p < q) (hgap : ∀ m, p < m → m < q → ¬ Nat.Prime m) :
    nthPrime (k + 1) = q
```

Each `hgap` obligation is an `interval_cases` over a gap of size < 10 in this range — milliseconds. The whole file (both bridge lemmas, the 26-entry table, *and* the certified range) compiles in **6.5–9.6 s**. **No numeral is asserted anywhere**: $p_1 = 2$ comes from Mathlib, and every later entry is *derived* from its predecessor.

Also proved and used: `count_prime_eq_of_no_prime` (no prime in $[a, b) \Rightarrow \pi$ constant across it), `nthPrime_succ_le_of_prime` (minimality), and the equivalence bridges `firoozbakht_iff_log`, `firoozbakht_iff_int`, `firoozbakht_iff_gap` — the machine-checked form of §1.2 and Lemma 1.

4.3 Both directional criteria, exact

Sufficient (verification side).

```
theorem firoozbakht_of_gap_lt {n : ℕ} (hn : 1 ≤ n)
    (h : (n : ℝ) * ((nthPrime (n + 1) : ℝ) - (nthPrime n : ℝ))
        < (nthPrime n : ℝ) * Real.log (nthPrime n)) :
    (nthPrime (n + 1) : ℝ) ^ ((1 : ℝ) / ((n : ℝ) + 1)) <
      (nthPrime n : ℝ) ^ ((1 : ℝ) / (n : ℝ))
```

i.e. $ng_n < p_n \log p_n \Rightarrow$ F at n. The proof is the single estimate $\log(p_{n+1}/p_n) \le p_{n+1}/p_n - 1$, which is exactly where the exact barrier degrades to its first-order surrogate. The hypothesis is therefore *strictly stronger* than the conjecture at n: **sufficient, never necessary.**

Necessary conditions for a refutation — two exact criteria.

```
theorem not_firoozbakht_of_barrier_le {n : ℕ} (hn : 1 ≤ n)
    (h : barrier n ≤ ((nthPrime (n + 1) : ℝ) - (nthPrime n : ℝ))) :
    ¬ ((nthPrime (n + 1) : ℝ) ^ ((1 : ℝ) / ((n : ℝ) + 1)) <
        (nthPrime n : ℝ) ^ ((1 : ℝ) / (n : ℝ)))
```

```
theorem not_firoozbakht_of_pow_le {n : ℕ} (hn : 1 ≤ n)
    (h : (nthPrime n) ^ (n + 1) ≤ (nthPrime (n + 1)) ^ n) :
    ¬ ((nthPrime (n + 1) : ℝ) ^ ((1 : ℝ) / ((n : ℝ) + 1)) <
        (nthPrime n : ℝ) ^ ((1 : ℝ) / (n : ℝ)))
```

The second is **pure ℕ arithmetic** — no real arithmetic, no `Real.log` — and is therefore the right target for a computational search. Note what it demands: the **index** $n = \pi(p_n)$ must be certified, not merely the gap. *A large gap alone is not a counterexample.*

Explicitly not admissible as a refutation: exhibiting a gap above $\log^2 p_n - \log p_n - 1$. That is the asymptotic surrogate, not B_n, and the difference is $O(1/\log p_n)$ — which is exactly the size of the margin being contested.

4.4 RT-12 discharge — hypothesis/conclusion separation

Per §2.4, for every declaration this paper names as a result:

declaration	hypotheses	conclusion	conclusion among hypotheses?
`firoozbakht_of_le`	$1 \le n, n \le 25$	F1 at n	**no**
`firoozbakht_gap_of_le`	$1 \le n, n \le 25$	$g_n < B_n$	**no**
`nthPrime_succ_eq`	$1 \le k$, `nthPrime k =` p, q prime, $p < q$, no prime strictly between	`nthPrime (k+1) = q`	**no**
`nthPrime_eq_1 … _26`	none	$p_i = $ numeral	**no**
`count_prime_eq_of_no_prime`	$a \le b$, no prime in $[a, b)$	π constant	**no**
`nthPrime_succ_le_of_prime`	$1 \le n$, q prime, $p_n < q$	$p_{n+1} \le q$	**no**
`firoozbakht_iff_log`	$1 \le n$	F1 $\iff$ F3	**no** (biconditional bridge)
`firoozbakht_iff_int`	$1 \le n$	F1 $\iff$ F2	**no**
`firoozbakht_iff_gap`	$1 \le n$	F1 $\iff$ $g_n < B_n$	**no**

declaration	hypotheses	conclusion	conclusion among hypotheses?
`firoozbakht_of_gap_lt`	$1 \le n$, $ng_n < p_n L_n$	F1 at n	**no**
`not_firoozbakht_of_barrier_le`	$1 \le n$, $B_n \le g_n$	¬F1 at n	**no**
`not_firoozbakht_of_pow_le`	$1 \le n$, $p_n^{n+1} \le p_{n+1}^n$	¬F1 at n	**no**
`nthPrime_succ_lt_two_mul`	$1 \le n$	$p_{n+1} < 2p_n$	**no**
`firoozbakht_of_two_pow_le`	$1 \le n$, $2^n \le p_n$	F1 at n	**no**

No declaration in this table takes Firoozbakht's conjecture — in any of its five forms — as a hypothesis. The two refutation criteria are genuine contrapositives; this was independently confirmed inside the kernel by typechecking

```
example (n : ℕ) (h1 : 1 ≤ n) (h2 : n ≤ 25)
    (hbad : (nthPrime n) ^ (n + 1) ≤ (nthPrime (n + 1)) ^ n) : False :=
  not_firoozbakht_of_pow_le h1 hbad (firoozbakht_of_le n h1 h2)
```

which wires the refutation criterion and the certified range together and shows the kernel agrees they are incompatible.

4.5 An external cross-check on the numerals

Lean checks that the proofs follow. It does not check that the *numerals* are the primes anyone means. Against an independent `sympy` sieve:

```
prime table p_1..p_26 matches sympy sieve: True
n in 1..25 failing p_{n+1}^n < p_n^(n+1): []
n with 2^n <= p_n (n < 40): [1]
n < 72849 where linearised criterion n*g_n < p_n*log p_n FAILS: count=2
first=[2, 4]
```

The third line is a genuinely useful fact: over the first ~73 000 indices the linearised sufficient criterion fails at only $n = 2$ and $n = 4$, **both inside the range already proved outright** by `firoozbakht_of_le`. So `firoozbakht_of_gap_lt` is not a lossy detour — and equally, it is **no easier than the target**.

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5. What was ESTABLISHED on paper

These are arguments, not certificates. Each carries its tier per §1.5.

5.1 The record locus, and the size of the margin

At the largest known maximal-gap locus [`kourbatov2015upper` Table 1, last row, **V0**]:

$$k = 49\,749\,629\,143\,526, \qquad p_k = 1\,693\,182\,318\,746\,371, \qquad g_k = 1132.$$

Recomputed here in 60-digit arithmetic from exact integer inputs (**[C]**):

quantity	recomputed	published	source
$\log p_k$	$35.0653861820133348\ldots$	—	—
$f_k = p_k^{1+1/k} - p_k$	$1193.41777829404\ldots$	1193.418	`kourbatov2015upper` Table 1 ✓
$\ell_k = \log^2 p_k - \log p_k$	$1194.51592191172\ldots$	1194.516	`kourbatov2015upper` Table 1 ✓
$R_{\text{csg}} = g_k / \log^2 p_k$	$0.9206385885574205\ldots$	0.9206	`primegaplist_faq` ✓
$f_k / \log^2 p_k$ (the fence, same normalisation)	$0.9705887446713\ldots$	—	**[C]**
shortfall f_k/g_k	$\mathbf{1.05425598789}\ldots$	—	**[C]**

Use 1.05426, sourced directly to the *published exact barrier* $f_k = 1193.418$. Not 1.055 (an earlier figure in this run, superseded), and not via any c_n interval — the exact barrier makes the explicit-bounds detour unnecessary. If an explicit-bounds route is wanted anyway it certifies 1.05424 ± 0.00005 (Dusart 2010 Props. 6.6/6.7, **V0**, validity $k \geq 688\,383$) — consistent, with the exact value inside it.

Tier: **L0/L1 + V0 + [C]**.

A caveat that is real, not decorative. The statement "the CSG ratio $R = g / \log^2 p$ has never exceeded 1" is **false as written**. It requires the restriction $p > 7$. Recomputed here: $R > 1$ at exactly three points — $p = 2$ ($R = 2.081369$), $p = 3$ (1.657071), $p = 7$ (1.056366) — and $R \leq 1$ for every other $p \leq 113$. The community source states the record *with* the restriction ("the largest value observed for any $p_1 > 7$"); any restatement must carry it.

5.2 Theorem A — an unconditional verified range at $H = 10^{20}$

> **Theorem A (established, not proved).** F holds for every n with $p_n < 10^{20}$.

The argument is complete and every step is either first-principles or one of exactly two external premises, both named and isolated:

premise	supplies	tier
P1 — Axler, *New bounds for the prime counting function*, the 3.83 member: $\pi(x) < x/\big(\log x - 1 - \frac{1}{\log x} - \frac{3.83}{\log^2 x}\big)$ for $x \geq 9.25$	the analytic criterion	**V0** · **L2** — see the locator fault below
P2 — the census of first-occurrence prime gaps is complete below 10^{20}, yielding the 85 known maximal gaps	the range	**V1** `[primegaplist_faq]` · **L3**: a community computational record, unrefereed, not machine-checked

Everything else — the criterion, its monotonicity, the covering lemma, the base case, the assembly — is L0. By the propagation rule the conclusion inherits its weakest premise: **Theorem A is L3/V1, and its weak link is computational data, not mathematics.**

The structural device is a **lever**. If the maximal-gap table is complete below H, with records (P_i, G_i), then for $P_i \leq p_n < P_{i+1}$ we have $g_n \leq G_i$ — a larger gap would be an unrecorded new record. Combined with a lower bound on B_n (from an *upper* bound on π, per §1.3), this turns ~80 inequalities into a verified range of $\approx 2.2 \times 10^{18}$ primes: 2.7×10^{16} **primes per inequality** [COMPUTED]. That ratio *is* the content of the phrase "verified up to 4×10^{18}". Nobody has evaluated the criterion at 10^{17} consecutive primes, and nobody will.

Two subsidiary findings:

- **The range is data-bound, not method-bound.** At the tightest of the 85 records the criterion is satisfied with 5.1% to spare [C]; H stops exactly where the census stops. The gain from 2^{64} (2019) to 10^{20} (today) is **not mathematical** — the same argument gives both; the census moved.
- **The relaxation costs nothing.** Using the unconditional criterion instead of the exact barrier loses 9.3×10^{-6} relative [C]. This is why the target closes at all.

A live citation fault on P1 — MAJOR-6, stated here rather than buried. The Axler result is cited throughout this corpus as **"Corollary 3.5"**, which is the **arXiv v3** numbering, against a bibliographic entry whose journal-of-record fields are *Integers* **16** (2016), A22. The published corrigendum numbers the affected result **Corollary 3.4, page 8**. A referee checking the version of record will **not** find the 3.83 bound at "Corollary 3.5". The premise is *believed* — the skeptic leg retrieved arXiv v3 and read the four-member family verbatim, confirming P1 at V0 — but the locator does not resolve in the version it is attributed to, and the corpus never establishes where the 3.83 member sits in the published numbering. **Theorem A must not be published until this locator is repaired.**

5.3 The exhaustive floor the project owns outright

Independently of any external premise: F verified exhaustively for all $p < 10^9$ — **50 847 533 consecutive pairs** — by sieve plus a *runtime-certified* floating-point error bound, in 9 s. Tier **L0 + [C]**. This is the only part of the range this work owns without inheriting anyone's computational claim. Below $p = 1223$ the check is exact integer comparison of p_{n+1}^n against p_n^{n+1}, with no arithmetic model at all.

Within that same floor, **FFM** — *if F first fails at* $n^\star$*, then* $g_{n^\star}$ *is a record gap* — is **certified, not assumed**, for all $n \leq 50\,847\,533$.

A vacuity note, applied here rather than inherited. On any range where F is *verified*, FFM is **vacuously true** and carries zero information: its antecedent is the negation of F. This paper therefore does **not** advertise "FFM proved on $[89, 10^{20})$" as a deliverable, because on exactly that range F is verified from exactly those premises. What is non-vacuous is the **certificate** $G_{n-1} < f_n$ on $[89, H)$ for every H to which the census is complete — strictly stronger than FFM-at-n and not implied by F's verification. The general FFM statement, for unbounded n, remains **open**.

5.4 The FFM dichotomy

The real deliverable of the FFM target is a structural theorem, not a range:

> **Theorem B (established).** Every known route to proving FFM passes either through F itself or through a statement strictly stronger than F.

Two elementary facts drive it. (a) F $\Rightarrow$ FFM vacuously. (b) Hence any *refutation* of FFM contains a refutation of F — to exhibit a counterexample to FFM one must exhibit the least index at which F fails, and in particular an index at which F fails. So

FFM is *logically weaker* than F, **and the weakness buys nothing**.

Tier **L0**.

---

6. What was REFUTED

Nothing about the conjecture. **Three routes** died, and the obituaries are quantitative.

6.1 The Riemann hypothesis route — dead, with numbers

> **Theorem C (established).** The strongest *published* RH-conditional prime-gap bound (Carneiro–Milinovich–Soundararajan 2019, constant $22/25$) decides Firoozbakht at **exactly three indices**, $n \in \{1, 2, 3\}$ — and at **exactly one**, $n = 3$ ($p = 5$), once the source's own validity restriction $p_n > 3$ is honoured. It is silent at every $n \geq 4$, forever.

> **Theorem D (established).** This is not a constant problem. For **every** $c > 0$, a bound $g_n \leq c\sqrt{p_n} \log p_n$ decides F on a finite initial segment only. Improving the constant moves a threshold; it never changes the verdict.

> **Theorem E (established).** The RH branch is **strictly dominated**. For every $c \geq 10^{-7}$ — seven orders of magnitude below the published constant, and six below $c = 1/2$, which the CMS authors name as the limit of their method — the usable set ends *inside* the range already verified **unconditionally**. RH contributes **zero indices** that unconditional verification does not already own.

Tier **L0 + [C]** for all three, resting on a **V0** reading of the CMS paper (but see the ledger caveat in §11.2).

The $\sqrt{x}$ shape is not laziness. Littlewood's unconditional $\Omega_\pm$ theorem forces any argument that bounds the prime-counting error pointwise and subtracts to work at that scale **regardless of RH**. (This explanatory proposition rests on a **V3** row for Littlewood 1914 and is stated as such; Theorems C–E do not use it.)

What was deliberately *not* proved: that RH does not *imply* Firoozbakht. That is a claim about RH's deductive closure and nothing here establishes it. The barrier proved is about the **shape of the bounds the known machinery emits**, over an explicitly enumerated surveyed set. That restraint is deliberate and is preserved in this paper.

One incidental positive. RH *does* have a use in this attack — on the **refutation** branch, certifying the index $n = \pi(p)$. It is on the wrong branch of the tree, and the unconditional bound was already comfortable there by a factor of ~1600.

6.2 The monotone-barrier route to FFM — dead, by an exact witness

The tempting proof of FFM routes through monotonicity of the exact barrier f_n. f_n **is not monotone**: it decreases at 57.88% of steps and sits below its own running maximum at 87.88% of indices, with an **exact-integer witness at** $n = 7, 8$ ($f_7 - f_8 = 0.0281326864429$, reproduced independently in 40-digit arithmetic).

The error is subtle and worth recording precisely: the **proxy** barrier $\ell_n = \log^2 p_n - \log p_n$ **is** monotone in the index; f_n is not. Substituting one for the other is the mistake, and it is what makes FFM "the price of exactness".

Tier **L0 + [C]**.

6.3 Unconditional analytic extension of the verified range — dead permanently

> **Theorem F (established).** No unconditional analytic result in the literature extends the verified range by a single prime, and none can: the best proven gap bound overshoots the barrier by a factor that **diverges** — 1.5×10^7 at 10^{20}, 6×10^{47} at 10^{100}.

The range is **data-bound, not method-bound**. Tier **L0 + [C]**.

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7. Computational evidence

Every number here is reproducible from the run directory; no RNG, no network.

7.1 What was computed, and on what tier

range	certified by	tier
$[2,\ 1223]$	exact integer comparison $p_{n+1}^{n} < p_{n}^{n+1}$	**L0** — no arithmetic model at all
$[2,\ 10^9)$	exhaustive sieve + runtime-certified error bound; 50 847 533 pairs, 9 s	**L0** — owned outright
$[89,\ 4 \times 10^{18})$	lever (gap table + Axler)	**L2**, 71 inequalities
$[89,\ 2^{64})$	same lever, longer table	**L2**, 76 inequalities
$[89,\ 10^{20})$	same lever, full b-file	**L2/L3** (community census), 80 inequalities
$[89,\ 1.857 \times 10^{19})$	second, independent lever needing no Axler bound	**L2**, 79 inequalities
beyond 10^{20}	nothing	open

No counterexample was found at any scale examined. No leg expected to find one, and none of these ranges is evidence about the conjecture.

7.2 Independent reproduction

An adversarial audit leg re-derived the load-bearing numbers in code written independently of the producing legs. Selected results:

- Re-sieved to 2×10^7 in independent NumPy: min $H_n = 1.196152422706632$ at $n = 2$; max $D_n = 0.5486599111467569$ at $n = 214$ ($p = 1307$) — **exact matches**.
- Recomputed the record-locus bracket: $1193.345367 \le 1193.417778 \le 1193.459723$ — **reproduces to every quoted digit**.
- Re-ran the 85-row lever: record #4 fails ($\underline{B}/G = 0.823$), #5 onward passes, tightest ratio 1.054246 at #64 — **matches**.
- Cross-checked two independently sourced maximal-gap tables (Wikipedia-sourced against OEIS b-files): **85/85 rows identical** in G, P and $\pi(P)$. *No producing leg had performed this cross-check.*
- Tested the unconditional criterion empirically at every index to 2×10^7: **0 violations**.

- Re-ran all seven verifier scripts: **all exit 0**, four producing byte-identical output.

7.3 A genuinely open computational obligation

Monotonicity of the lower barrier bound $\underline{B}$ is used and **not proved**. It was tested on a grid ~500× denser than any producing leg used — 300 001 points over $[10, 10^{22}]$ — with **0 non-monotone steps**. The gap remains a gap. Recorded as an open obligation, not as a result.

7.4 A correction to this work's own opening framing

The attack opened with "the refutation branch is the live branch; the shortfall is 5.5%". That framing was **partially demoted** during the run, and this paper carries the demotion rather than the headline:

- the shortfall figure is arithmetically right (corrected: 1.05426, §5.1) but is **not L0** as originally derived — that derivation traversed an unclosed literature node; it is L0 *now*, via the published exact barrier;
- the sufficient criterion the headline leans on is **silent at** $n = 2$ **and** $n = 4$ — the *tightest known cases* of the conjecture. F is true there by the exact predicate ($25 < 27$; $14641 < 16807$), but the criterion does not know it;
- of two **pre-registered** demotion criteria, one fired and one did not. The naive probabilistic model predicts 6.72 counterexamples below the verified frontier where 0 are observed — over-predicting by $6.7\times$; the fitted crossover height is 10^{30}–10^{51}, not 10^{21}.

The honest reading: the refutation branch keeps its heuristic motivation — Granville's $\limsup_x \max_{p_n \le x}(p_{n+1} - p_n)/\log^2 x \gtrsim 2e^{-\gamma} \approx 1.12292$ is not excluded, and this lim sup is genuinely unknown — and **loses its quantitative one**. Search is not affordable at any of the fitted heights.

The crossover-height figure itself carries a live caveat: it is a **30-decade extrapolation from a model fit**, and the model tag did not survive into the decision that used it. It is reported here as a model output, never as a property of the primes.

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8. Evidence gate: BLOCKED

The gate is fail-closed and it closed. This section reports the verdict as the gate returned it.

leg	requirement	verdict
LOOP	verdict file present, well-formed, names the live round	**PASS** — live round = 1
KERNEL	`lake build` exit 0 **and** grep-clean of `sorry`/`axiom`	**FAIL** — exit 0 ✅ but one live `sorry`; 7 declarations carry `sorryAx`
SKEPTIC	fault report exists **and** zero residual BLOCKERs	**FAIL** — 2 live BLOCKERs
CORPUS	red-team corpus present **and** coverage report non-empty	**PASS** — 19 entries / 15 categories

FAIL, not DEGRADED. DEGRADED is available only when no formal backend was attempted. Lean *was* run, *did* build, and *did not* discharge the `sorry`. Calling that DEGRADED would convert "we tried and the conjecture is still open" into "we could not try" — a different and false statement.

8.1 The two live BLOCKERs

The audit found **no mathematical error in any theorem**. Both BLOCKERs are claims the corpus makes *about its own trustworthiness*, and both are false. That is exactly the class of fault a gate exists to stop.

id	fault
BLOCKER-1	A provenance record states that 44 rows of the load-bearing maximal-gap table were independently recomputed. **Thirty were.** A 47% overstatement, sitting in the provenance record of the corpus's single weakest premise (P2 of Theorem A).
BLOCKER-2	The claim *"the verified range itself excludes $C \geq 1.1736$"* is an **invalid inference**: a finite range cannot exclude a value of a $\limsup$. It is a property of a model fit, read as a statement about the tail.

Neither needs new mathematics. One is a wrong row count; one is a sentence that must be deleted or restated. **This paper does not rely on either claim**: §7.2 reports only counts the audit itself verified, and §7.4 states the lim sup as unknown.

8.2 Seven MAJOR faults, carried forward

Dominated by citation and cross-leg consistency: a locator absent from the version of record (**MAJOR-6**, disclosed in §5.2); two mutually incompatible verification tiers for the same source (**MAJOR-7**); one leg crediting a sibling with a premise the sibling had retracted (**MAJOR-4**); a vacuous conclusion sold as a headline verdict (**MAJOR-3**, corrected in §5.3); a docstring carrying a formally retracted claim (**MAJOR-5**); an undated superlative survey claim inside a V0 row (**MAJOR-8**, see §11.2); and the 30-decade extrapolation feeding a budget decision (**MAJOR-9**, caveated in §7.4).

Of these, **MAJOR-3 and MAJOR-9 are corrected in this paper**; **MAJOR-6 and MAJOR-8 are disclosed at the point of use**; **MAJOR-4, MAJOR-5 and MAJOR-7 concern upstream artifacts this paper does not rely on.**

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9. Limitations — stated in full, not summarised

1. **The conjecture is open.** It is not proved and not refuted here. Nothing in this paper is evidence about its truth.
2. **No finite range is evidence.** Neither the kernel-checked $n \le 25$, nor the exhaustive $p < 10^9$, nor Theorem A's 10^{20}. The whole difficulty is in the tail.
3. **Theorem A inherits an unrefereed computational premise (P2, L3/V1)** and a **faulted citation locator (P1, MAJOR-6)**. It should be read as *medium* confidence, with the weak link on the data side.
4. **The citation audit has not run.** No citation here has been independently re-fetched and re-read by an auditor. §11 traces each to a ledger row; that is provenance, not clearance.
5. **Six citations used in §6.1 sit on ledger rows that were proposed but never merged** into the authoritative ledger. §11.2 lists them explicitly.
6. **Three named gaps remain unsourced or unusable**: the four-term expansion $B_n = L^2 - L - 1 - 3/L - 13/L^2 + O(L^{-3})$ is **unsourced** — the published theorem gives only the $+o(1)$ form, and this paper therefore does **not** use the lower-order terms; the Baker–Harman–Pintz threshold x_0 is **ineffective as published**, so 0.525 must never be quoted as an explicit

unconditional bound without that caveat; and no live URL exists for the canonical Nicely gap tables (all return HTTP 404 as of 2026-07-24) — cite the community project instead.

7. **The monotonicity of $\underline{B}$ (§7.3) is used and unproved.**
8. **Only one attack round ran** (a configured cap, not a failure — the executor was verified capable). There is therefore no shrink-versus-churn signal: the list of `sorry`-carrying declarations entered at 7 and left at 7, **identical, not churned**. The six corollaries were never independent targets — they inherit from the head.
9. **`lake build` was not re-executed by this leg.** The build record (exit 0, 8253 jobs) is read as the kernel leg's own transcript.
10. **The delivery posture is staged.** §0.2.

9.1 What must close before release

In the order the evidence imposes:

1. Close **BLOCKER-1** (correct the row count) and **BLOCKER-2** (delete or restate the invalid inference), and unwind any decision resting on the latter.
2. Repair the **MAJOR-6** locator: establish where the 3.83 member sits in the *published* Axler numbering, or re-attribute P1 to arXiv v3 explicitly in every place it appears.
3. Merge or drop the six **proposed-but-unmerged ledger rows** of §11.2. Theorems C–E cannot ship on unmerged rows.
4. Run **`citation-gate`**, then **`editorial-verdict`**.

Steps 1–4 are corpus-honesty and citation work, not mathematics. **None of them will close the kernel leg**: that closes only if the conjecture is proved, which is the open problem itself.

9.2 The honest forecast

More rounds of the same shape would very likely keep producing scaffolding without moving the head. The obstruction is not a missing lemma that a re-read would surface — it is the **categorical gap between polylogarithmic and power-of-p gap bounds** (§1.4). If further rounds are run, their value is in closing the corpus faults and hardening the criteria, **not** in expecting the `sorry` to fall.

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10. What this paper does not claim

- It does not claim the conjecture is true, false, likely, or unlikely. The community's heuristic expectation (from Granville's revised Cramér model) is that it is **false**; that is recorded as a heuristic and is used as a premise nowhere.
- It does not claim citation clearance (§9, limitation 4; §11).
- It does not claim that RH fails to imply Firoozbakht (§6.1).
- It does not claim any verified range is evidence.
- It does not claim FFM on any range as a substantive result (§5.3).
- It does not use the unsourced lower-order terms of the B_n expansion.
- It does not claim "proved" for anything outside §4.

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11. Citation trace

Every citation in this paper resolves to a row in the run's authoritative source ledger, `source-ledger/source-ledger.md`. Section numbers below are that document's.

11.1 Merged ledger rows used in this paper

citekey	ledger §	V	locator relied on	used in
`firoozbakht1982`	§3.1	V1	attribution + date only	§1.1, §0.3
`rivera2002conj30`	§3.1	V1	page body	§1.1
`ribenboim2004little`	§3.1	V2	p. 185	§1.1
`visser2019verifying`	§3.1, §3.2, §3.8	**Vo**	Conj. 3 eq. (2.4); Abstract; §4	§1.3, §4.1, §7.1
`kourbatov2015verification`	§3.2	V1	Abstract	§4.1, §7.1
`kourbatov2015upper`	§3.3, §3.4	**Vo + C**	Table 1 last row; Thm 5; §3	§1.3, §5.1
`kourbatov2015corrigendum`	§3.4	**Vo**	full text	§11.3
`primegaplist_faq`	§3.2, §3.3	V1 + C	FAQ body	§5.1, §5.2, §7.1
`axler2014newbounds`	§3.5	**Vo**	Cor. 3.5 (arXiv v3) — **see**	§5.2

citekey	ledger §	V	locator relied on	used in
			MAJOR-6	
`axler2016corrigendum`	§3.5	V2	via Kourbatov corrigendum	§5.2, §11.3
`dusart2010estimates`	§3.5	**V0 + C**	Props. 6.6, 6.7	§5.1
`granville1995harald`	§3.6	**V0**	display following eq. (20); eq. (14)	§7.4
`cramer1936order`	§3.6	V2	block quotation in Granville	§11.3
`bhp2001difference`	§3.6	V2	main theorem	§1.4; §9, limitation 6
`fgkmt2018long`	§3.7	V1	Abstract	§11.3

11.2 Proposed but unmerged rows — a live obligation

The RH-route results of §6.1 rest on citations that a downstream leg **proposed** for the ledger and that were **never merged** into it. They are recorded here with the verification level the proposing leg claimed, and they are **not cleared**:

citekey	claimed V	statement relied on	status
`cms2019fourier` — Carneiro, Milinovich, Soundararajan, *Fourier optimization and prime gaps*, Comment. Math. Helv. **94** (2019), 533–568; arXiv:1708.04122	**V0** (claimed; quotations independently re-fetched and confirmed exact by the audit leg)	on RH, $g_n \le \frac{22}{25}\sqrt{p_n}\log p_n$ for $p_n > 3$; the method's limiting constant is $\frac{1}{2}$	**UNMERGI**
`littlewood1914`	**V3**	$\psi(x) - x = \Omega_{\pm}(\sqrt{x}\log\log\log x)$	**UNMERGI + V3** — usec only for the explanatory remark in §6 which is flagged as su
`dudek2015riemann`	V1	earlier RH-conditional constant	**UNMERGI** lineage only

citekey	claimed V	statement relied on	status
`leenosal2025sharper`	**V0**	sharper RH error bounds	**UNMERGE** not load-bearing here
`schoenfeld1976sharper`	V3 → V2 (tier disputed, **MAJOR-7**)	classical RH bound	**UNMERGE not used in this paper**
`montgomery1973pair`	V3	pair-correlation consequence	**UNMERGE not used in this paper**

Consequence, stated plainly. Theorems C, D and E of §6.1 are **established** mathematics resting on a citation that is not yet in the authoritative ledger. They must not be released until `cms2019fourier` is merged at V0 and `littlewood1914` is either promoted or the explanatory remark is restated as conditional on it. This is item 3 of §9.1.

A further caveat on the same family (**MAJOR-8**): the phrase *"the strongest published RH-conditional gap bound"* is a **superlative survey claim**, and no leg documents a literature sweep after 2019 — one was attempted and blocked by rate limiting. §6.1 therefore says "the strongest *published* bound" as the producing leg's assessment, not as a verified fact about the literature as of 2026.

11.3 Citation hygiene notes that must survive into any restatement

Five errors are easy to make in this literature, and the ledger pins each:

1. **The $b = 1.17$ sufficient condition has a corrected validity range.** Kourbatov's Theorem 3 originally rested on Axler's bound at "$x \geq 5.43$"; Axler's own corrigendum moved that to $x \geq 2\,634\,800\,823$, and Kourbatov published a corrigendum rewriting the proof. Any citation of "$b = 1.17 \Rightarrow$ Firoozbakht" that does not carry $p_k \geq 2\,634\,800\,823$ **plus** the unconditional check on $[29,\ 2\,634\,800\,823]$ is citing a withdrawn proof. Cite arXiv:1506.03042**v4**.
2. **The verification frontier is 2^{64}, not 4×10^{18}.** 4×10^{18} is Kourbatov's 2015 exhaustive-sieve range; Visser 2019 covers $p < 2^{64} \approx 1.844 \times 10^{19}$, for Firoozbakht *and* the two strictly stronger Nicholson and Farhadian conjectures. The community census reaches 10^{20} (§5.2, P2).

3. **Explicit bounds here are Axler's, not Rosser–Schoenfeld's or Dusart's.** Kourbatov's chain uses Axler. Writing "Rosser–Schoenfeld / Dusart" while reproducing that derivation is mis-citing. (Dusart 2010 *is* used, at V0, but for the p_k bracket of §5.1 — a different role.)
4. **The long-gaps exponent is** $(\log\log\log X)^1$**, not squared.** The 2018 record improves the 2016 one by exactly one iterated log in that denominator. This is the easiest citation error in the family. Either way the bound is $\ll \log^2 X$ — too small by a factor $\sim \log X/\log\log X$ to touch Firoozbakht.
5. **Cramér stated the** $\limsup = 1$ **relation about his random model, and only *suggested* the transfer to the primes.** Any sentence beginning "Cramér conjectured" must carry that hedge. And "Cramér 1920/21" is a mis-citation: the RH gap bound is Cramér 1920, with a distinct companion paper the same year.

Also recorded as **absences**, which is part of the provenance: there is **no primary document by Firoozbakht** (the standard reference is literally "*(1982), unpublished*"), so "Firoozbakht proposed, in 1982" is supportable and "Firoozbakht proved/published" is not; the canonical Nicely gap tables are **dead links**; and **no proof, disproof, or accepted counterexample of the conjecture was found anywhere in the literature.**

The bibliography accompanying this paper is `references.bib` (31 merged citekeys, plus a clearly separated addendum block for §11.2's unmerged rows).

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12. Reproduction

All artifacts of the run live under the run directory. The load-bearing checks:

```
# formal layer
cd <run>/lean-probe/lean && lake build          # exit 0; one `sorry`
warning, at Statement.lean
                                                # Audit.lean prints
axioms for 31 declarations

# recomputation layer (each deterministic, no RNG, no network)
python3 <run>/source-ledger/verify_ledger.py    # ledger rows
python3 <run>/proof-attempt__0/verify_pa0.py    # FFM numerics
python3 <run>/proof-attempt__1/verify_rh.py     # RH usable sets
python3 <run>/proof-attempt__2/verify_pa2.py    # the 85-row lever
python3 <run>/notebooks__2/verify-findings.py   # the verified-range
```

```
lever
python3 <run>/concept-cards/verify_cards.py     # concept-card
invariants
python3 <run>/skeptic/verify_faults.py          # the audit's own re-
derivations
python3 <run>/synthesize/verify_synthesis.py    # 56/56 checks, exit 0
```

The synthesis verifier is deliberately a **tripwire**: it asserts that the two BLOCKERs are still live. If a future session closes them, those checks **fail** — which is the intended behaviour, not a regression.

---

13. The one-paragraph version

The primes walk along a road, and every step is a gap. Firoozbakht drew a fence at height roughly $(\log p)^2$ and said the primes never jump it — not once, not ever, all the way to infinity. This work did not decide that. What it did was survey the ground carefully. It showed the fence is *exactly* the right description of the conjecture, not an approximation. It proved — inside a kernel, in the literal form — that the primes stay under it for the first 25 steps, and established by argument that they do so for every prime below 10^{20}, resting on two named external premises. It showed that the Riemann hypothesis, the biggest hammer anyone would reach for, settles the question for the prime 5 and for no other. It showed that no unconditional method in the literature can extend the verified range by a single prime, and that the shortfall **diverges** rather than closing. And it built the two exact criteria that a future proof and a future refutation would each need. Then its own auditor found two places where the corpus was overstating how carefully it had checked itself, and the gate closed red on that. **The mathematics in this corpus is sound. Its self-description was not, in two places. The conjecture is still open.**

---

*Emitted by the `write-paper` leg (`edit-20260724-9e06`) of the `math-attack` polymer, run node. External attribution: **Noogram**. Firoozbakht's conjecture is **open** — not proved, not refuted. Evidence-gate status: **BLOCKED**. Citation clearance: **not claimed, not audited**. Delivery posture: **staged**.*